

University of Cape Town
University Examinations – May/June 2023
Department of Mathematics and Applied Mathematics
MAM3000W

Module 3MS: Metric Spaces

Duration : 2 hours

Full Marks: 40

Please answer Question 1 and 2 in one answer book, and Questions 3 to 5 in another.

1. For x, y real numbers, let

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ |x| + |y| & \text{if } x \neq y \end{cases}$$

- (a) Prove carefully that d is a metric on $\mathbb{R}$.
- (b) Find the open ball $B(2023; 2030)$ in $(\mathbb{R}, d)$.
- (c) Give an example of a sequence (x_n) such that $x_n \rightarrow 7$ in $(\mathbb{R}, d)$. No proof required.
- (d) Let $\mathbb{Q}$ denote the set of rational numbers, find $\text{int}(\mathbb{R} \setminus \mathbb{Q})$ in $(\mathbb{R}, d)$. Justify your answer.

[13]

2. (a) Prove that every open ball in a metric space is an open set.
- (b) Prove or disprove: If X is a connected metric space and $f : X \rightarrow Y$ is a continuous surjective function, then Y is connected.

[6]

Please change to a new answer book now.

3. (a) Prove or disprove: Let $(F_i)_{i \in I}$ be a family of closed subsets of a metric space, then $F = \bigcap_{i \in I} F_i$ is a closed subset of the metric space, that is, arbitrary intersection of closed sets is closed.
- (b) Prove or disprove: If $(A_i)_{i \in I}$ be a family of subsets of a metric space, then $\overline{\bigcap_{i \in I} A_i} = \bigcap_{i \in I} \overline{A_i}$.

[5]

4. Read the argument below, and then answer the questions that follow.

Cantor's Intersection Theorem Let (X, d) be a metric space. Then (X, d) is complete iff this condition holds:

For any decreasing sequence (A_n) of non-empty subsets of X with $\text{diam } A_n \rightarrow 0$,

$$\bigcap_{n \in \mathbb{N}} \overline{A_n} \neq \emptyset.$$

Proof $(\implies)$ Omitted. (1)
 $(\impliedby)$ Begin with a Cauchy sequence (x_n) in (X, d) . (2)
 For each $n \in \mathbb{N}$, let $T_n = \{x_k \mid k \geq n, k \in \mathbb{N}\}$. (3)
 Then (T_n) is a decreasing sequence of sets. (4)
 Let $k_n = \text{diam } T_n$. Then $k_n \rightarrow 0$. (5)
 So, by assumption, there exists $x \in \bigcap_{n \in \mathbb{N}} \overline{T_n}$. (6)
 We now show that $x_n \rightarrow x$. (7)
 To do that, begin with $\epsilon > 0$. (8)
 Choose N such that $\text{diam } T_N < \epsilon/2$. (9)
 Take $b \in B(x; \epsilon/2) \cap T_N$. (10)
 Then $n \geq N \implies d(x_n, x) < \epsilon$. (11)

- (a) Justify the claim in line (4).
- (b) Prove that $\text{diam } T_n \rightarrow 0$, as claimed in line (5).
- (c) How do we know the N mentioned in line (9) exists?
- (d) How do we know that $B(x, \epsilon/2) \cap T_N \neq \emptyset$, as used in line (10)?
- (e) Show that line (11) follows from the previous working.
- (f) Somewhere in this argument, the fact that Cauchy sequences are bounded is used. You need not prove this result; only explain where in this proof it is used. [10]

5. (a) Let X and Y be sets and $f : X \rightarrow Y$ be a function.
 Let $\mathcal{T} = \{A \subseteq X \mid f^{-1}(f(A)) = A\}$.
 Prove that $\mathcal{T}$ is a topology on X .
- (b) Prove or disprove: Let $\mathcal{T} = \{A \subseteq \mathbb{R} \mid A' \text{ is finite}\} \cup \{\emptyset\}$ (this $\mathcal{T}$ is the cofinite topology on $\mathbb{R}$, no need to verify that it is a topology on $\mathbb{R}$). $\mathcal{T}$ is induced by a metric. [6]

THE END